

PROJECT REPORT

AI & ML Internship Task 7

Project Title: Exploratory Data Analysis (EDA) on Automobile Dataset

Submitted By:
MODI DHRUV BHAVESHBHAI

Internship Program:
AI & ML Internship
Edtech Solution

Edutech Solution

AI & ML INTERNSHIP

Task 7: Exploratory Data Analysis (EDA)

Tools:

- Python (Pandas, Matplotlib, Seaborn)

Dataset:

- "Automobile Dataset" or "Ecommerce Sales Data"

Hints / Mini Guide:

1. Perform Univariate Analysis (Distributions of individual variables).
2. Perform Bivariate Analysis (Correlations between variables).
3. Use Multivariate Analysis (Relationships between three or more variables).
4. Generate heatmaps for feature correlations.
5. Visualize data using Box plots to detect outliers.
6. Create Pairplots to observe group-wise data trends.
7. Summarize key insights and actionable findings.

Deliverables:

- Jupyter Notebook containing all visualizations and code
- A report highlighting the top 5 insights from the data

Final Outcome:

Intern builds strong analytical thinking and visual storytelling skills.

Interview Questions Related To Above Task:

- What is the importance of EDA in the Data Science lifecycle?
- Explain the difference between Univariate and Bivariate analysis.
- How do you handle skewed data discovered during EDA?
- What does a correlation coefficient of -1 indicate?
- How do you decide which visualization to use for categorical vs. numerical data?

Table of Contents

1. Cover Page
2. Declaration
3. Acknowledgement
4. Table of Contents
5. Introduction
 - 5.1 What is Exploratory Data Analysis
 - 5.2 Importance of EDA in Data Science
 - 5.3 Types of Data Analysis
 - 5.4 Automobile Industry Data Applications
6. Problem Statement
7. Project Objectives
8. Dataset Description
 - 8.1 Dataset Overview
 - 8.2 Features Description Table
9. Tools and Technologies Used
10. Methodology
 - 10.1 Dataset Loading and Data Inspection
 - 10.2 Data Cleaning
 - 10.3 Univariate Analysis
 - 10.4 Outlier Detection
 - 10.5 Bivariate Analysis
 - 10.6 Correlation Analysis
 - 10.7 Multivariate Analysis
11. Implementation and Results
12. Challenges Faced

13. Learning Outcomes

14. Conclusion

15. References

5. Introduction

5.1 What is Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an approach to analyzing data sets to summarize their main characteristics, often using statistical graphics and other data visualization methods. It is the critical first step in any data investigation, allowing analysts to discover patterns, spot anomalies, test hypotheses, and check assumptions with the help of summary statistics and graphical representations.

5.2 Importance of EDA in Data Science

EDA is important because it ensures that the data is ready for machine learning models. Without EDA, a data scientist might build models on faulty assumptions or dirty data. EDA helps in understanding the distribution of data, identifying missing values, detecting outliers, and understanding the relationships between different variables, which directly informs feature selection and engineering.

5.3 Types of Data Analysis

There are three primary forms of EDA implemented in this report:

1. Univariate Analysis: Examining a single variable to understand its distribution and central tendencies.
2. Bivariate Analysis: Analyzing the relationship between two distinct variables to identify correlations.
3. Multivariate Analysis: Observing the interactions between three or more variables simultaneously to detect complex patterns.

5.4 Automobile Industry Data Applications

In the context of the automobile industry, EDA allows engineers and business analysts to understand how physical vehicle parameters (like weight, dimensions, and engine size) practically interact with performance metrics (like fuel efficiency) and risk profiles. This data drives decision-making in vehicle design and pricing strategies.

6. Problem Statement

Raw datasets contain a mixture of numerical, categorical, and missing data points. The objective of this project is to process the "Automobile Dataset", diagnose its structural and statistical anomalies, and systematically apply visualization techniques to extract actionable business and design insights. The challenge lies in accurately interpreting visual data to form logical conclusions regarding vehicle performance and physical attributes.

7. Project Objectives

1. Perform Univariate Analysis to understand individual feature distributions.
2. Visualize data using Box plots to detect and analyze outliers.
3. Perform Bivariate Analysis to find direct relationships between two features.
4. Generate Heatmaps to map the correlation coefficients of all numerical variables.
5. Create Pairplots to observe group-wise data trends.
6. Summarize key R&D and business insights based on the visual findings.

8. Dataset Description

8.1 Dataset Overview

The Automobile dataset comprises various physical and mechanical attributes of vehicles, along with their insurance risk ratings and fuel efficiency metrics. It contains both categorical (make, fuel-type, body-style) and continuous numerical data (length, width, curb-weight, engine-size, mpg).

8.2 Features Description Table

Key numerical features analyzed include:

- symboling: Insurance risk rating (+3 is high risk, -3 is safe).
- wheel-base, length, width, height: Dimensional physical attributes of the car.
- curb-weight: Total mass of the vehicle without passengers.
- engine-size: Volumetric capacity of the engine.
- compression-ratio: Ratio of the volume of the cylinder.
- city-mpg & highway-mpg: Fuel efficiency metrics.

9. Tools and Technologies Used

- Programming Language: Python 3
- Environment: Jupyter Notebook / Google Colab
- Data Manipulation: Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn
- Documentation: Microsoft Word

10. Methodology

10.1 Dataset Loading and Data Inspection

Loading the CSV using Pandas and inspecting its shape, datatypes, and initial rows using functions like `.head()` and `.info()`.

10.2 Data Cleaning

Handling missing values via median imputation for numerical data and mode imputation for categorical data. Dropping duplicate rows to ensure data integrity.

10.3 Univariate Analysis

Plotting Histograms and Kernel Density Estimates (KDE) for numerical columns to understand data skewness and distribution.

10.4 Outlier Detection

Using Seaborn Boxplots to map the Interquartile Range (IQR) and identify extreme statistical deviations.

10.5 Bivariate Analysis

Creating scatter plots between specific variables (e.g., Symboling vs Wheel-base) to trace linear or non-linear trends.

10.6 Correlation Analysis

Generating a correlation matrix and displaying it using a Seaborn Heatmap to rapidly identify multicollinearity.

10.7 Multivariate Analysis

Utilizing pairplots to view the holistic interactions among all continuous variables simultaneously.

11. Implementation and Results

11.1 Distribution of Compression Ratio

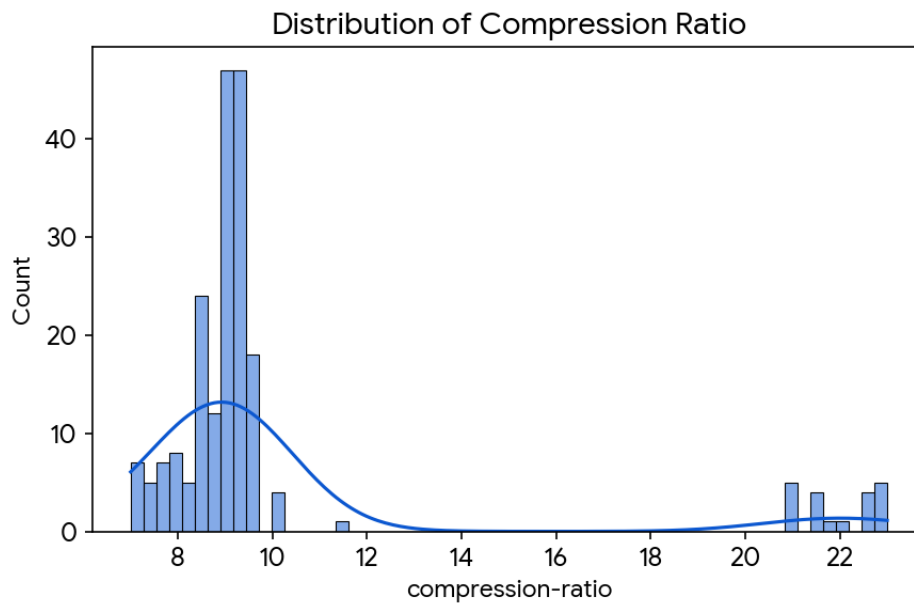


Figure 10.1: Distribution of Compression Ratio

Purpose of Visualization: To observe the spread, central tendency, and skewness of the compression ratio.

Observation: Highly concentrated around 8 to 10. Secondary small cluster beyond 20.

Interpretation: Primary cluster represents standard gasoline engines; secondary cluster indicates diesel engines.

Business/Data Science Insight: The data is strongly bimodal. Predictive models must segment gas and diesel vehicles.

11.2 Distribution of City MPG

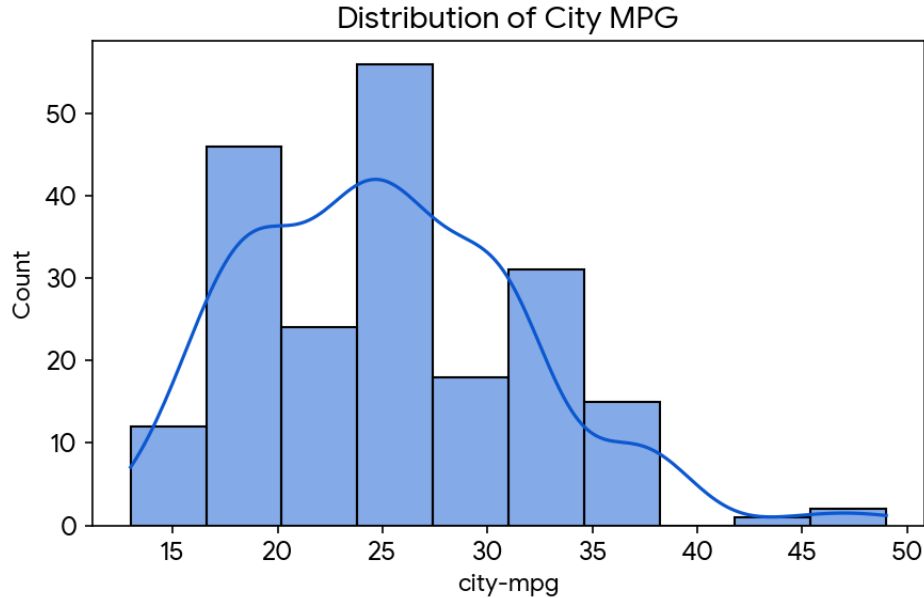


Figure 10.2: Distribution of City MPG

Purpose of Visualization: To understand typical fuel efficiency in urban driving conditions.

Observation: Right-skewed, peaking heavily around 15-25 MPG.

Interpretation: Most vehicles are standard consumer cars with average economy. High-efficiency cars are rare.

Business/Data Science Insight: Achieving a City MPG above 30 places the vehicle in a competitive, niche segment.

11.3 Distribution of Highway MPG

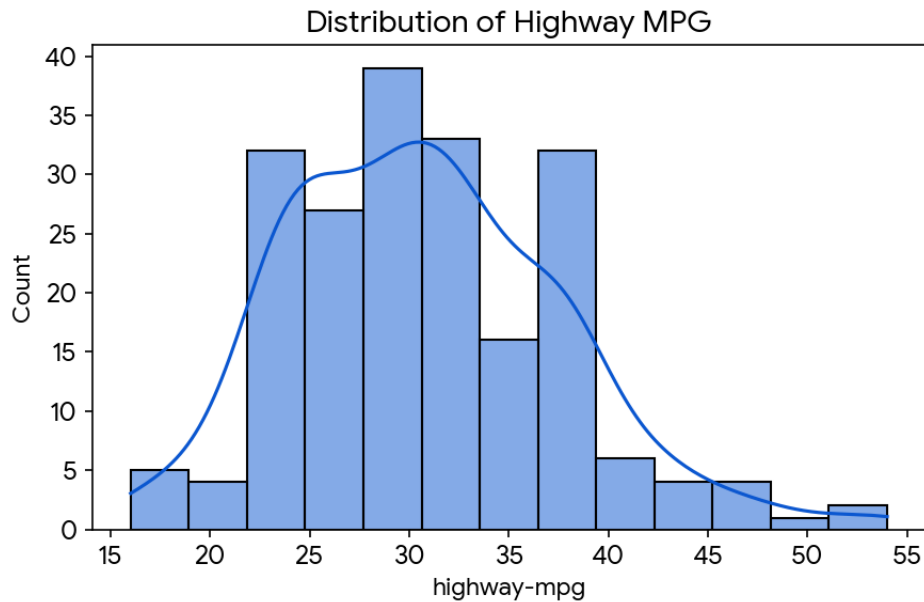


Figure 10.3: Distribution of Highway MPG

Purpose of Visualization: To evaluate fuel efficiency during steady-state highway driving.

Observation: Shifted higher, mostly between 25 and 35 MPG.

Interpretation: Vehicles perform better on highways due to reduced stop-and-go mechanics.

Business/Data Science Insight: Validates physical engineering expectations. True high-efficiency vehicles are outliers.

11.4 Boxplot of Compression Ratio

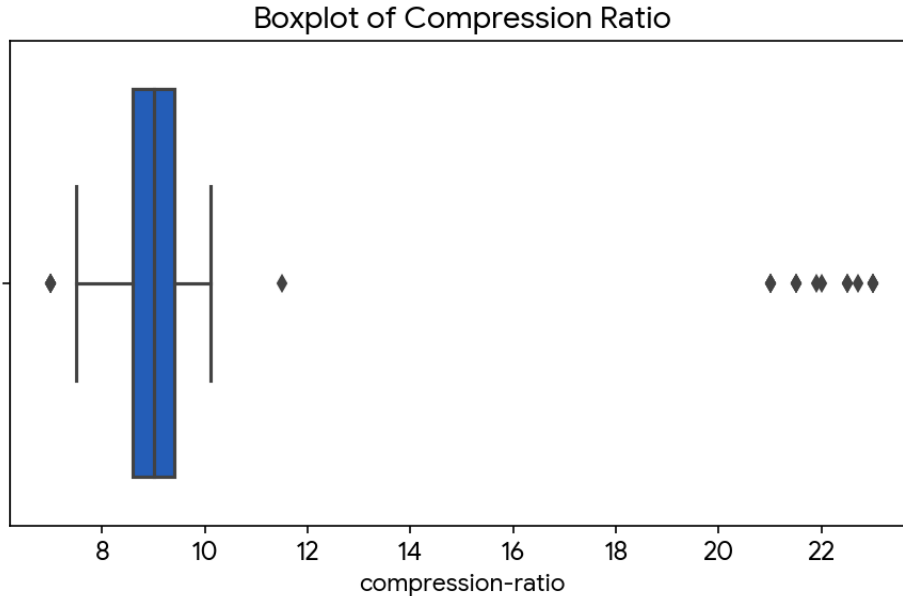


Figure 10.4: Boxplot of Compression Ratio

Purpose of Visualization: To detect statistical outliers and understand the interquartile range (IQR).

Observation: Narrow box centered below 10. Distinct outliers on the higher end (>20).

Interpretation: Because diesel engines are rare in this dataset, they are statistically classified as outliers.

Business/Data Science Insight: An analyst must not remove these outliers, as they represent valid physical characteristics.

11.5 Boxplot of City MPG

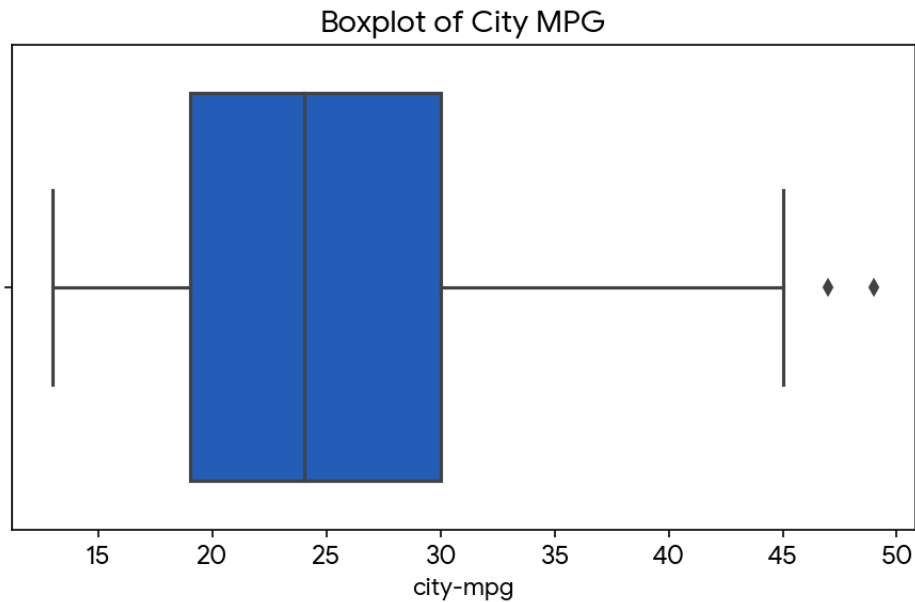


Figure 10.5: Boxplot of City MPG

Purpose of Visualization: To visually separate the bulk of data from exceptionally high performing vehicles.

Observation: Outliers extend past the upper whisker near 45-50 MPG.

Interpretation: These upper outliers represent specialized economy vehicles.

Business/Data Science Insight: If building a predictive model for standard vehicles, these ultra-efficient cars might skew results.

11.6 Boxplot of Highway MPG

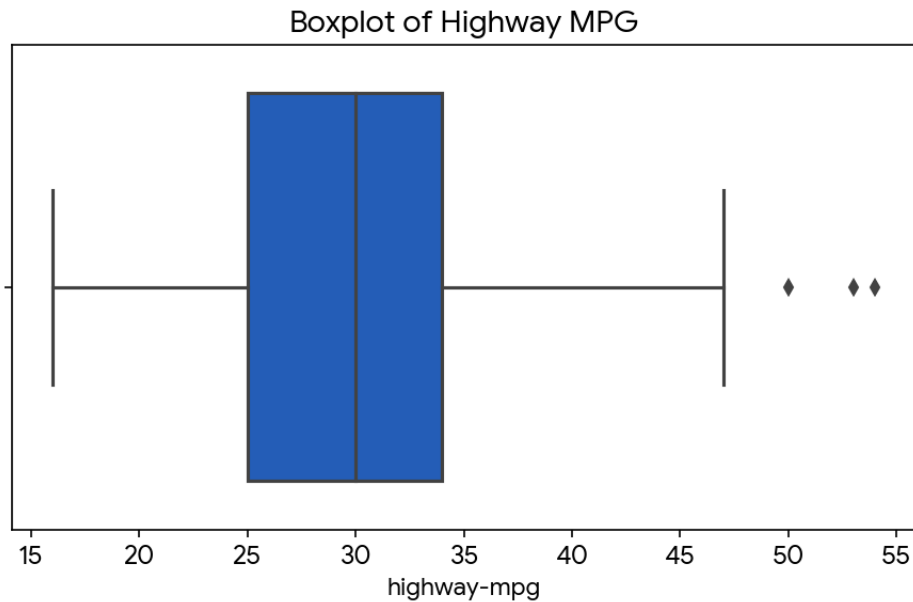


Figure 10.6: Boxplot of Highway MPG

Purpose of Visualization: To view variance and extreme values in highway fuel efficiency.

Observation: IQR spans 25-34 MPG. Outliers exist above 50 MPG.

Interpretation: A small fraction is highly optimized for fuel economy.

Business/Data Science Insight: Identifying specs of these outliers yields valuable R&D data.

11.7 Scatter Plot (Symboling vs Wheel-base)

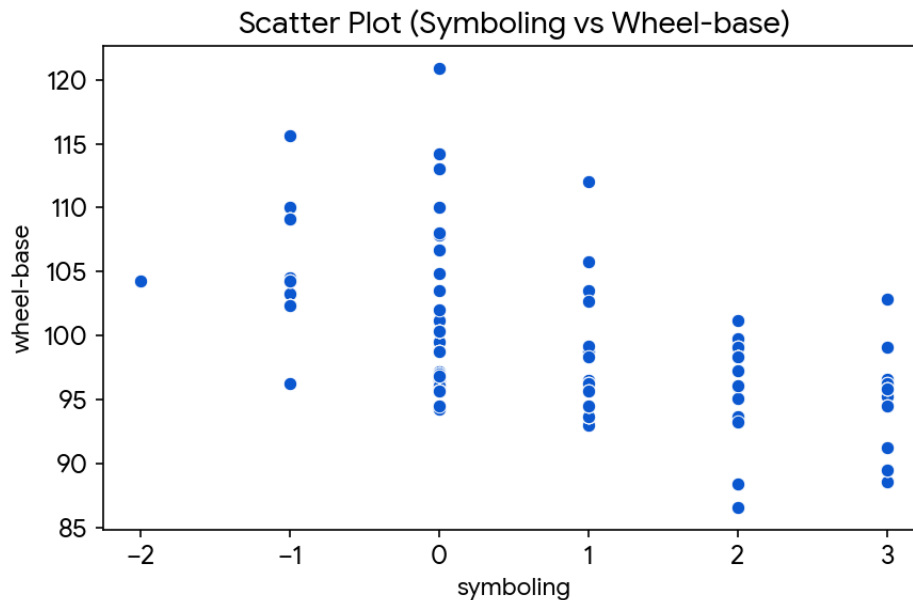


Figure 10.7: Scatter Plot (Symboling vs Wheel-base)

Purpose of Visualization: To determine if insurance risk correlates with a major physical vehicle dimension.

Observation: Noticeable negative trend. Higher symboling (+2, +3) has shorter wheel-base.

Interpretation: Shorter wheel-bases are sportier cars (higher risk). Longer wheel-bases are larger sedans (safer).

Business/Data Science Insight: Insurance actuaries can utilize physical dimensions as a direct proxy for establishing base risk profiles.

11.8 Correlation Heatmap

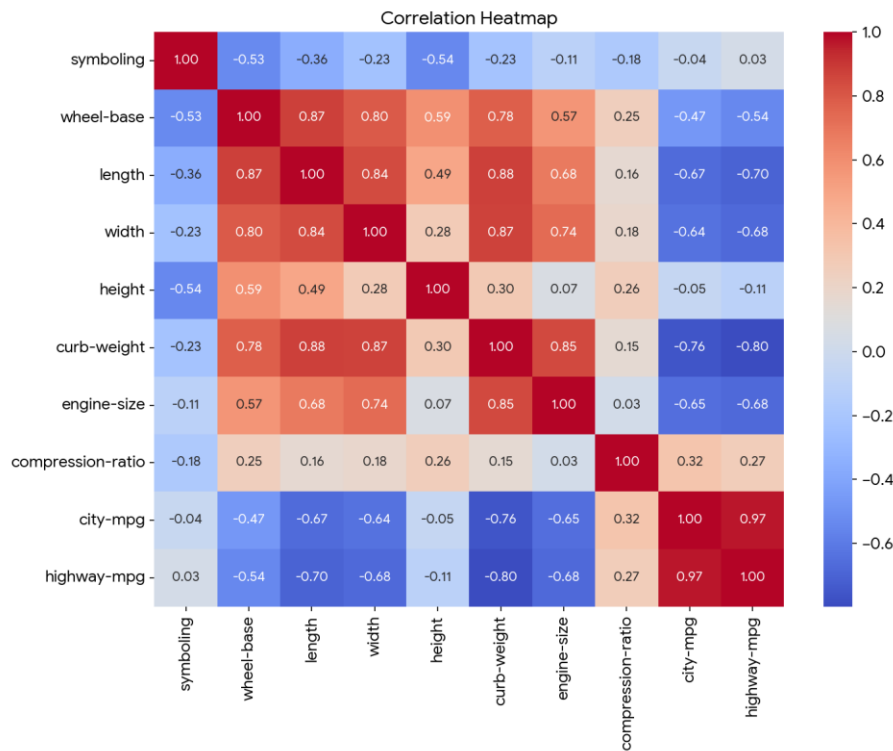


Figure 10.8: Correlation Heatmap

Purpose of Visualization: To calculate and display Pearson correlation coefficients.

Observation: Dark red patches indicate strong positive correlation (e.g., length and weight). Dark blue indicates negative (weight and MPG).

Interpretation: Key mechanical dimensions scale together, while size inversely affects efficiency.

Business/Data Science Insight: The heatmap allows for immediate feature selection by identifying redundant variables.

11.9 Pairplot of Numerical Features

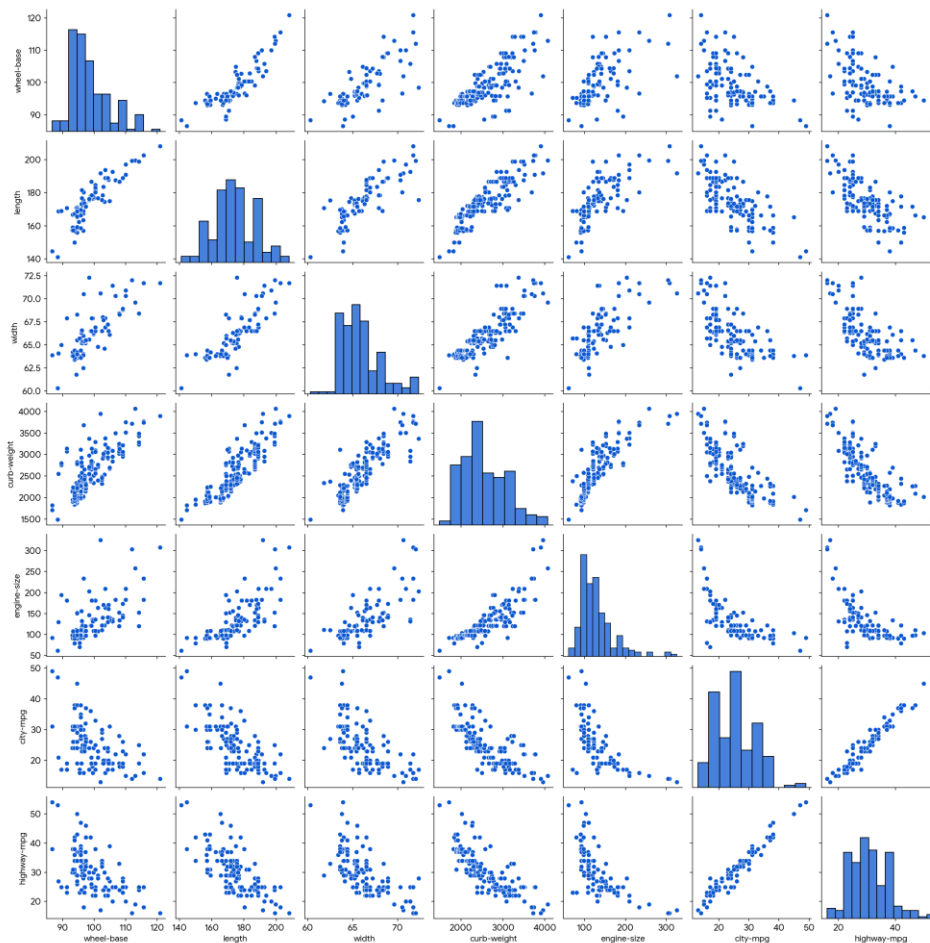


Figure 10.9: Pairplot of Numerical Features

Purpose of Visualization: To observe group-wise data trends and distributions.

Observation: Linear clusters between weight/dimensions. Sweeping downward curves when comparing weight against MPG.

Interpretation: As mass increases, MPG decreases non-linearly.

Business/Data Science Insight: To increase fuel efficiency, engineers must prioritize lightweighting the chassis.

12. Challenges Faced

1. Dealing with Skewed Data: Many continuous features, such as City MPG, exhibited significant skewness, which complicated standard distribution interpretations.
2. Interpreting Outliers: Distinguishing between data entry errors and valid mechanical anomalies (e.g., diesel engines acting as outliers in compression ratio).

3. Handling Missing Values: Making statistical decisions on whether to drop rows or impute them using median/mode to maintain data integrity.

13. Learning Outcomes

1. Mastered Pandas for data ingestion, missing value imputation, and dataset cleaning.
2. Developed proficiency in Matplotlib and Seaborn for generating aesthetic, informative visualizations.
3. Gained the ability to interpret correlation matrices and translate mathematical coefficients into real-world business logic.
4. Learned to recognize the importance of domain knowledge (automotive engineering) when interpreting statistical outliers.
5. Built strong analytical storytelling skills, structuring findings logically from data ingestion to actionable insight.

14. Conclusion

The Exploratory Data Analysis of the Automobile dataset successfully decoded the complex mechanical and performance relationships inherent in vehicle design. The visual evidence proved that physical dimensions are inextricably linked to vehicle mass, which acts as the primary antagonist to fuel efficiency. Careful outlier analysis revealed that anomalies often represent distinct engineering categories rather than errors. This task reinforced the absolute necessity of EDA in Data Science, ensuring that subsequent machine learning algorithms are grounded in physical logic and mathematical reality.

15. References

[1] Automobile Dataset

<https://www.kaggle.com/datasets/toramky/automobile-dataset>

[2] Google Colab Notebook

https://colab.research.google.com/drive/1Dzrh7EupG_FMEAyvNVKhzGMj7YNIBifb?usp=sharing

[3] GitHub Repository

<https://github.com/dbmodi2311/Data-Collection-Data-Source-Identification>

[4] Pandas Documentation

<https://pandas.pydata.org/docs/>

[5] Seaborn Documentation

<https://seaborn.pydata.org/>